

Abstract

Pacific herring (*Clupea pallasii*) spawning creates a concentrated coastal resource pulse. We tested whether coastal birds were reported more often and in higher positive counts near recorded spawn, whether associations concentrated in spawn and egg-availability windows, whether focal responses exceeded a prespecified specificity panel, and whether patterns recurred across regions. We linked complete eBird checklists to event-level Fisheries and Oceans Canada spawn records and used mixed models adjusted for effort, observer, year, location, and event structure. Analyses included 217,200 Strait of Georgia and 8,584 West Coast Vancouver Island checklist events. All 49 support-qualified species were displayed regardless of prior literature prominence or coefficient direction. Species-level associations were heterogeneous: SoG detection, SoG conditional-count, WCVI detection, and WCVI conditional-count models yielded 36/49, 34/49, 42/49, and 38/48 positive completed coefficients, respectively. Common shore and sea birds showed positive, negative, and mixed regional patterns; Surf Scoter and Short-billed Gull had positive coefficients for both response components in both regions. Event-window coefficients did not form a single resource-pulse trajectory. Matched sensitivities generally preserved direction, and shifted-exposure placebos were null after Benjamini-Hochberg adjustment. The Strait of Georgia specificity panel was non-null for Gadwall (beta = 0.234, 95% CI 0.132–0.337; $q = 7.28 \times 10^{-6}$) and Northern Shoveler (beta = 0.628, 0.527–0.729; $q = 6.56 \times 10^{-34}$). Spawning therefore left a detectable but heterogeneous and not uniformly taxon-specific signature in eligible submitted checklists, supporting a qualified resource-pulse interpretation rather than a causal or population-level effect.

Keywords: Pacific herring; eBird; community science; coastal birds; resource pulse; mixed models; falsification analysis